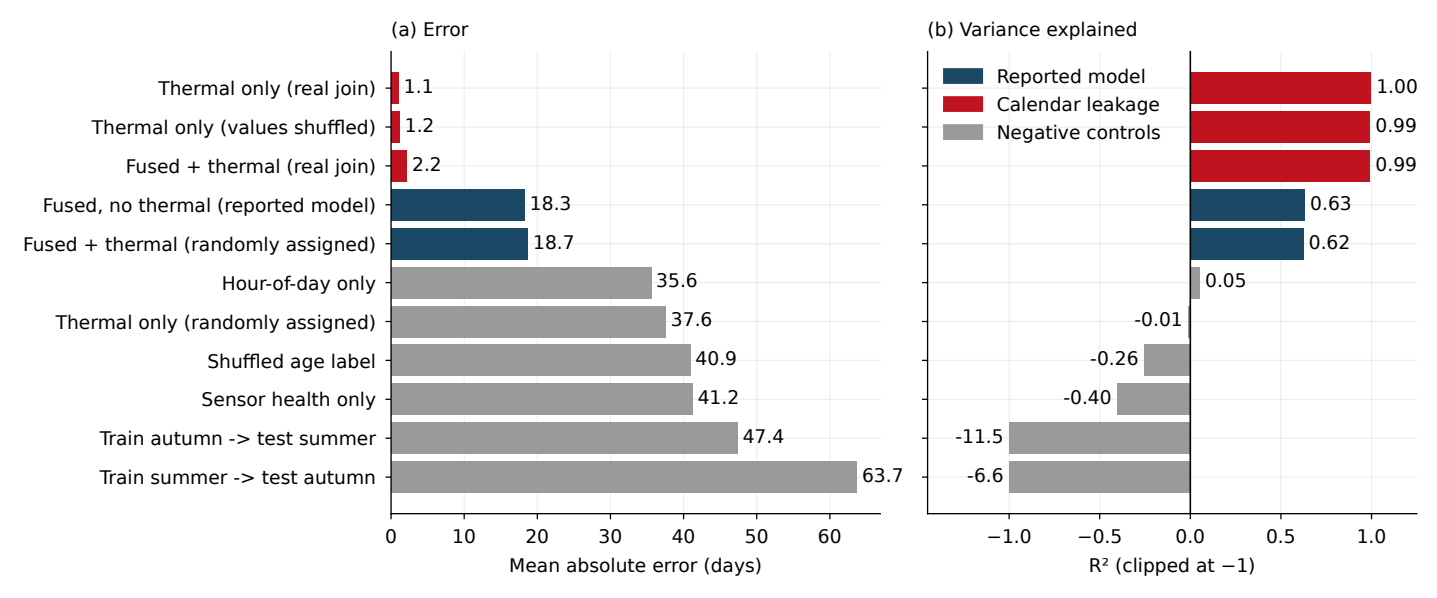
(a) Error
Thermal only (real join)
Thermal only (values shuffled)
Fused + thermal (real join)
Fused, no thermal (reported model)
Fused + thermal (randomly assigned)
Hour-of-day only
Thermal only (randomly assigned)
Shuffled age label
Sensor health only
Train autumn -> test summer
Train summer -> test autumn
1.1
1.2
2.2
18.3
18.7
35.6
37.6
40.9
41.2
47.4
63.7
0
10
20
30
40
50
60
Mean absolute error (days)
(b) Variance explained
Reported model
Calendar leakage
Negative controls
1.00
0.99
0.99
0.63
0.62
0.05
-0.01
-0.26
-0.40
-11.5
-6.6
−1.0
−0.5
0.0
0.5
1.0
R² (clipped at −1)